

Branch: MCA (Data Science) Kargil	Semester: 2
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Experiment 8

Experiment Aim:

To apply the concept of Stored Procedures in database operations in order to perform tasks like insertion, updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

Tools used: PostgreSQL

Objectives:

- **Procedure Implementation:** To understand how to create and use stored procedures for database operations.
- **Code Reusability:** To reduce redundancy by encapsulating logic inside procedures.

- **Data Manipulation:** To perform INSERT, UPDATE, DELETE operations using procedures.
- **Data Retrieval:** To understand the use of functions for returning result sets.

Theory:

Stored Procedures are precompiled SQL code blocks stored in the database that can be executed repeatedly. They improve performance, enhance security, and promote modular programming. PostgreSQL supports procedures (for actions) and functions (for returning data).

Experiment Steps:

Step 1: Creating Table Query:

```
CREATE TABLE Student (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    marks INT  
);
```

Step 2: Inserting Sample Data Query:

```
INSERT INTO Student VALUES  
(1, 'Aarav', 75),  
(2, 'Diya', 82),  
(3, 'Kabir', 68),
```

(4, 'Ananya', 90),
(5, 'Rohan', 77),
(6, 'Sneha', 85),
(7, 'Vivaan', 92),
(8, 'Megha', 80);

	student_id [PK] integer	name character varying (50)	marks integer
1	1	Aarav	75
2	2	Diya	82
3	3	Kabir	68
4	4	Ananya	90
5	5	Rohan	77
6	6	Sneha	85
7	7	Vivaan	92
8	8	Megha	80

Step 3: Stored Procedure for INSERT Query:

```
CREATE OR REPLACE PROCEDURE add_student(
    p_id INT,
    p_name VARCHAR,
    p_marks INT)
LANGUAGE plpgsql
AS $$
BEGIN
    INSERT INTO Student(student_id, name, marks)
    VALUES (p_id, p_name, p_marks);
END;
$$;
```

Step 4: Stored Procedure for UPDATE Query:

```
CREATE OR REPLACE PROCEDURE update_marks(  
    p_id INT,  
    p_marks INT  
)  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    UPDATE Student  
    SET marks = p_marks  
    WHERE student_id = p_id;  
END;  
$$;
```

Step 5: Stored Procedure for DELETE Query:

```
CREATE OR REPLACE PROCEDURE delete_student(  
    p_id INT  
)  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    DELETE FROM Student  
    WHERE student_id = p_id;  
END;  
$$;
```

Step 6: Function for SELECT (Retrieval)

Query:

```
CREATE OR REPLACE FUNCTION get_all_students()
RETURNS TABLE(student_id INT, name VARCHAR, marks INT)
LANGUAGE plpgsql
AS $$
BEGIN
    RETURN QUERY
    SELECT * FROM Student;
END;
$$;
```

Step 7: Calling Procedures and Function

Query:

-- Insert

```
CALL add_student(9, 'Ishaan', 88);
```

-- Update

```
CALL update_marks(2, 90);
```

-- Delete

```
CALL delete_student(3);
```

-- Select

```
SELECT * FROM get_all_students();
```

Output:

	student_id integer	name character varying	marks integer
1	1	Aarav	75
2	4	Ananya	90
3	5	Rohan	77
4	6	Sneha	85
5	7	Vivaan	92
6	8	Megha	80
7	9	Ishaan	88
8	2	Diya	90

Outcomes:

- I will be able to create and execute stored procedures.
- I will understand the difference between procedures and functions.
- I will perform CRUD operations using procedures.
- I will retrieve data using functions in PostgreSQL.